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Math Methods-I (PHY421)

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Homework-6

1. Obtain the Fourier series representation of the following periodic function:

$$\begin{aligned} f(x) &= -C, \quad -L < x < 0 \\ &= +C, \quad 0 < x < L, \end{aligned}$$

where C is a real constant. Using any standard mathematical software, plot the Fourier series representation of $f(x)$ (in units of C) vs x/L over $-5 \leq x/L \leq +5$ for three different values of the upper limit of the series summation: $n_{\max} = 50, 100, 200$. Analyse the plots around the point of discontinuity and then you will see the Gibb's phenomenon.

2. Show that the Fourier sine representation of the function $f(x) = x(\pi - x)$ with $0 \leq x \leq \pi$ is

$$f(x) = \frac{8}{\pi} \sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{(2n+1)^3}.$$

Plot the function $f(x)$ and its Fourier sine representation (keeping few terms in the series) over $0 \leq x \leq \pi$ on the same frame.

3. (a) Show that the Fourier series expansion of $f(t) = \cos(\omega_0 t)$ (ω_0 is not an integer) in the interval $(-\pi, \pi)$ is

$$\cos(\omega_0 t) = \frac{\sin(\pi\omega_0)}{\pi\omega_0} \left[1 + 2\omega_0^2 \sum_{n=1}^{\infty} \frac{(-1)^n \cos(nt)}{\omega_0^2 - n^2} \right].$$

- (b) Obtain Fourier transform of $f(t)$ (here ω_0 can be integer also):

$$F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt.$$

- (c) Plot the Fourier series expansion of $f(t)$ and sketch Fourier transform of $f(t)$ i.e. $F(\omega)$.

4. The Fourier series expansion of a periodic function $f(x) = x$ with a period $2L$ is given by

$$f(x) = L \left[1 - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{\sin(n\pi x/L)}{n} \right].$$

Using any standard mathematical software, plot the Fourier series expansion of $f(x)$ (in units of L) vs x/L over $-6 \leq x/L \leq +6$ for three different values of the upper limit of the series summation: $n_{\max} = 10, 50, 100$. Analyse the plots around the point of discontinuity and then you will see the Gibb's phenomenon.

5. (a) Defining $f(x) = e^{\pm x}$ are periodic functions with the fundamental interval $(-\pi < x < \pi)$. Show that the Fourier series of the periodic function $f(x)$ ($-\pi < x < \pi$) is given by

$$e^{\pm x} = \frac{\sinh(\pi)}{\pi} \left[1 + 2 \sum_{n=1}^{\infty} (-1)^n \frac{\cos(nx) \mp n \sin(nx)}{1 + n^2} \right].$$

- (b) Defining $\sinh(x)$ and $\cosh(x)$ are periodic functions in the interval $-\pi < x < \pi$. Obtain the Fourier series of $\sinh(x)$ and $\cosh(x)$.

6. A function $f(x)$ is expanded in an exponential Fourier series

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \quad \text{with} \quad c_n = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-inx} dx.$$

- (a) If $f(x)$ is real, what restriction is imposed on the coefficients c_n ?
 (b) If $f(-x) = f(x)$, what restriction is imposed on the coefficients c_n ?
 (c) If $f(-x) = -f(x)$, what restriction is imposed on the coefficients c_n ?
 (d) Assuming that the Fourier expansion of a periodic function $f(x)$ ($0 < x < 2\pi$) is convergent, show that the Parseval's identity has the form given by

$$\frac{1}{2\pi} \int_0^{2\pi} [f(x)]^2 dx = c_0^2 + 2 \sum_{n=1}^{\infty} |c_n|^2.$$

7. Obtain the Fourier series form of the periodic function $f(x) = x^4$ ($-\pi < x < \pi$). Using this result, show that

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

8. Solve the following inhomogeneous differential equation:

$$\omega \frac{df(\theta)}{d\theta} + f(\theta) = a \sin \theta + b \cos(2\theta).$$

Here ω, a, b are real constant and $f(\theta)$ is a periodic function with period 2π .

9. **Dirac comb:** It is a periodic, infinite array of δ -functions with the fundamental interval $(-L/2, L/2)$ i.e. $\delta(x) = \sum_{n=-\infty}^{\infty} \delta(x - nL)$. The Dirac comb model potential is popularly used to study band structure in solid state systems.

Show that the Fourier series representation of $\delta(x)$ is

$$\delta(x) = \frac{1}{L} \sum_{n=-\infty}^{\infty} e^{\pm i 2\pi n x / L} = \frac{1}{L} \left[1 + 2 \sum_{n=1}^{\infty} \cos(2\pi n x / L) \right].$$

Plot this function using any standard mathematical software.

10. a) Show that the Dirac delta function $\delta(x - a)$ expanded in a Fourier sine series in the interval $(0, L)$ ($0 < a < L$) is given by

$$\delta(x - a) = \frac{2}{L} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi a}{L}\right) \sin\left(\frac{n\pi x}{L}\right).$$

- b) A string is clamped at both ends $x = 0$ and $x = L$. Assuming small-amplitude vibrations, the amplitude $u(x, t)$ satisfies the wave equation

$$\frac{\partial^2 u(x, t)}{\partial x^2} = \frac{1}{c_s^2} \frac{\partial^2 u(x, t)}{\partial t^2},$$

where c_s is the velocity of the wave propagation along the string. The string is set in vibration by a sharp blow at $x = a$. Hence the initial conditions are

$$u(x, 0) = 0 \quad \text{and} \quad \left. \frac{\partial u(x, t)}{\partial t} \right|_{t=0} = Lv_0 \delta(x - a).$$

Solve the wave equation subject to these initial conditions.

11. A string, clamped at $x = 0$ and $x = L$, is vibrating freely. Its motion is described by the wave equation

$$\frac{\partial^2 u(x, t)}{\partial x^2} = \frac{1}{c_s^2} \frac{\partial^2 u(x, t)}{\partial t^2},$$

where c_s is the velocity of the wave propagation along the string. Assume a Fourier series of the form

$$u(x, t) = \sum_{n=1}^{\infty} b_n(t) \sin\left(\frac{n\pi x}{L}\right).$$

The initial conditions are $u(x, 0) = f(x)$ and $\left.\frac{\partial u(x, t)}{\partial t}\right|_{t=0} = g(x)$. Determine the coefficients $b_n(t)$.

12. The one-dimensional heat equation is given by

$$D \frac{\partial^2 u(x, t)}{\partial x^2} = \frac{\partial u(x, t)}{\partial t},$$

where D is the diffusivity. The boundary conditions are $u(0, t) = 0$ and $u(L, t) = 0$. Show that the most general solution of the heat equation with the given boundary conditions is

$$u(x, t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{L}\right) e^{-n^2 \pi^2 D t / L}.$$

Determine the Fourier coefficients a_n if the initial condition is $u(x, 0) = f(x)$.

13. The position of an underdamped oscillator is given by

$$x(t) = e^{-\gamma t} \cos(\omega_0 t) \theta(t),$$

where γ is the damping factor, ω_0 is the undamped oscillator frequency and $\theta(t)$ is the unit step function.

(a) Show that the Fourier transform of $x(t)$ is

$$x(\omega) = \frac{1}{i\sqrt{2\pi}} \left[\frac{\omega - i\gamma}{(\omega - i\gamma)^2 - \omega_0^2} \right].$$

(b) Plot $x(t)$ vs t and $|x(\omega)|^2$ vs ω for $\omega_0 = 10$ and $\gamma = 1$. Compare the plots.

14. Obtain Fourier transform of the following functions:

(a) The ground and first excited state wave functions of a hydrogen atom

$$\psi_{100}(\mathbf{r}) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}, \quad \psi_{200}(\mathbf{r}) = \frac{1}{\sqrt{32\pi a_0^3}} \left(2 - \frac{r}{a_0}\right) e^{-r/(2a_0)}.$$

Here a_0 is the Bohr radius and $r = \sqrt{x^2 + y^2 + z^2}$.

(b) The ground and first excited state wave functions of 1D harmonic oscillator

$$\phi_0(x) = \frac{1}{\sqrt{\sqrt{\pi} a_h}} e^{-x^2/(2a_h^2)}, \quad \phi_1(x) = \sqrt{\frac{2}{\sqrt{\pi} a_h^3}} x e^{-x^2/(2a_h^2)},$$

where $a_h = \sqrt{\hbar/(m\omega)}$ is the oscillator length.

- (c) The ground state wave function of a shifted harmonic oscillator

$$\phi_0(x) = \frac{1}{\sqrt{\sqrt{\pi}a_h}} e^{-(x-x_0)^2/(2a_h^2)},$$

where x_0 is a constant.

- (d) The Coulomb potential in 3D:

$$V(\mathbf{r}) = \frac{C}{r},$$

where $C = 1/(4\pi\epsilon_0)$.

- (e) The Gaussian potential in 3D:

$$V(\mathbf{r}) = V_0 e^{-\frac{r^2}{4a_h^2}}.$$

- (f)

$$V(\mathbf{r}) = V_0 a_h^5 \nabla^2 \delta^3(\mathbf{r}).$$

15. Calculate the following inverse Fourier transformation of the Fermi occupation number $n_{\mathbf{k}} = \Theta(k_f - |\mathbf{k}|)$ at $T = 0$:

$$n(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int_{V_{\mathbf{k}}} e^{i\mathbf{k}\cdot\mathbf{r}} n_{\mathbf{k}} d^3k,$$

where $V_{\mathbf{k}}$ is the volume of a sphere with radius k_f in $\mathbf{k}$ -space. Here $\mathbf{k}$ and $\mathbf{r}$ are the three-dimensional vectors. Plot $n(\mathbf{r})$ vs $k_f r$.

Obtain approximate expressions of $n(\mathbf{r})$ for $r \rightarrow 0$ and $r \rightarrow \infty$ limits and sketch them.

Repeat the same calculations for 2D systems.

Answer for 3D case:

$$n(\mathbf{r}) \propto \frac{\sin(k_f r) - (k_f r) \cos(k_f r)}{r^3}.$$

For small r : $n(y) \simeq 1/3 - y^2/(30)\dots$ and for large r : $n(y) \simeq \frac{\cos y}{y^2}$ with $y = k_f r$.

This kind of integrals appear while calculating Hartree-Fock energy.

16. The drift-diffusion equation is given by

$$\frac{\partial n(x, t)}{\partial t} = D \frac{\partial^2 n(x, t)}{\partial x^2} - v_d \frac{\partial n(x, t)}{\partial x},$$

where $n(x, t)$ is the particle density, D is the diffusivity or diffusion constant and v_d is the drift velocity.

- (a) Using the Fourier transform method, solve for $n(x, t)$ of the drift-diffusion equation for a given initial condition $n(x, 0) = C_0 \delta(x)$ with C_0 being a constant. Sketch $n(x, t)$ at different times.

17. The diffusion equation is given by

$$\frac{\partial n(x, t)}{\partial t} = D \frac{\partial^2 n(x, t)}{\partial x^2},$$

where $n(x, t)$ is the particle density and D is the diffusivity or diffusion constant. One can construct a typical length scale of a diffusive system as $l_D = \sqrt{2Dt} \sim \sqrt{t}$. Keep in mind that the length scale goes as $\sqrt{t}$ and it is not a constant. This equation can be solved by assuming the following ansatz:

$$n(x, t) = \frac{N_0}{l_D} f\left(\frac{x}{l_D}\right).$$

Here, N_0 is the total particle number in the system and it is conserved i.e. at any given time t , $\int_{-\infty}^{\infty} n(x, t) dx = N_0$. Introducing a dimensionless variable $\xi = x/\sqrt{2Dt} = x/l_D$. Therefore, $f(\xi)$ is a function of one variable ξ .

a) Show that $\int_{-\infty}^{\infty} f(\xi) d\xi = 1$.

b) Show that the partial diffusion equation reduces to the following ordinary differential equation:

$$\left[\frac{d^2}{d\xi^2} + \xi \frac{d}{d\xi} + 1 \right] f(\xi) = 0.$$

Assuming initial condition is $f(x, t = 0) = N_0 \delta(x)$. First solve the ordinary differential equation for $f(\xi)$ and then show that the solution of the partial differential equation is

$$n(x, t) = \frac{N_0}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}}.$$

Guess what will be the solution of the drift-diffusion equation

$$\frac{\partial n(x, t)}{\partial t} = D \frac{\partial^2 n(x, t)}{\partial x^2} - v_d \frac{\partial n(x, t)}{\partial x},$$

where v_d is the drift velocity.

18. Using Fourier transform method, solve the following inhomogeneous differential equation

$$\left[\frac{d^2}{dx^2} + k_0^2 \right] f(x) = -e^{-x^2},$$

where k_0 is a constant and $f(x)$ vanishes as $|x| \rightarrow \infty$.

Answer:

$$f(x) = \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{e^{-k^2/4}}{k^2 - k_0^2} e^{ikx} dk.$$

19. Using the Parseval's identity, show that

$$\int_{-\infty}^{\infty} \frac{d\omega}{(w^2 + a^2)^2} = \frac{\pi}{2a^3}.$$